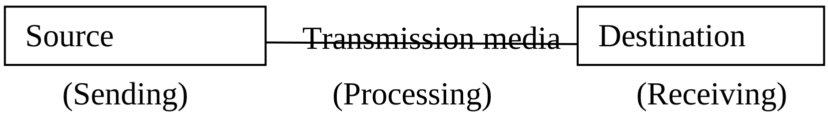
∆∆OPTICAL FIBER COMMUNICATION

Communication

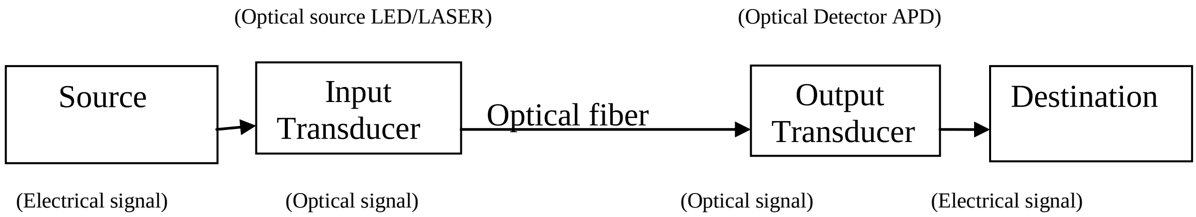
Sending, processing and receiving the information from source to destination is called as communication.



Optical Fiber Communication

Sending, processing and receiving the information from source to destination through the transmission media as optical fiber is called as optical fiber communication.

Optical fiber allows optical (light) signals only, for purpose of transmission. It doesn't allow the electrical signals directly. If the source generates electrical signal, first it should converted into optical signal using transducer. Transducer is nothing but converting one form of energy into another form of energy. Then the optical signal is transmitted through optical fiber. Finally, it is received by the destination. We have to get the original signal as electrical, so again the transducer is used at the receiver to convert the optical signal into electrical signal.



Optical sources LED/LASER acts as input transducer at transmitter section, which converts electrical signal into optical one.

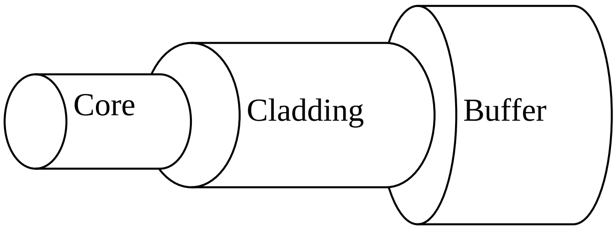
Optical detector APD (Avalanche photo diode) acts as output transducer at receiver section, which converts optical signal into electrical signal.

Optical fiber construction

Optical fiber is a hair-thin, flexible, transparent medium. There are made with glasses and plastics. The basic material used to make glass fiber is silica (SiO_2).

Optical fiber consists of three layers

1. Core
2. Cladding
3. Buffer coating jacket



1. The light signal has to be travel within the core of the optical fiber, it should not get refracted into the cladding.

2. Cladding is used to prevent the leakage of light signal. It allows the light signal to be propagated within the core.

3. Outer jacket is used to prevent the fiber from scratches and any other type of damages.

Optical fiber works based on the principle of the Total Internal Reflection (TIR).

Total Internal Reflection

Consider two mediums having different Refractive Index n_1 and n_2.

Refractive Index is defined as the ratio of speed of light in vacuum to the speed of light in medium.

$$R.I = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium (air, glass, water)}}$$

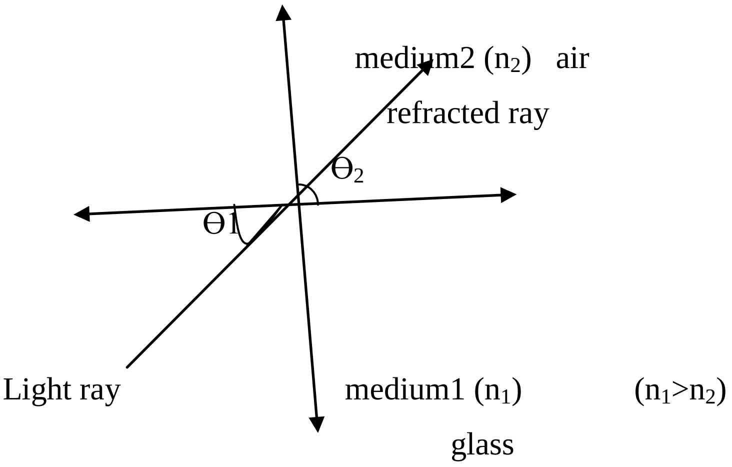
For air, R.I=1

For glass, R.I=1.5

For water, R.I=1.33

Case-1:

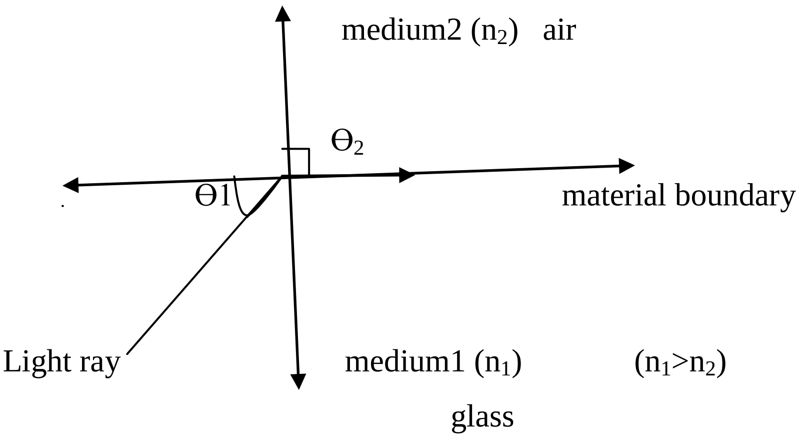
($\Theta_1 < \Theta_2$)



When the incident angle (Θ_1) is less than the critical angle (Θ_c), the light signal will refracted into the second medium. That means the light signal enters into the cladding, represents loss of signal.

Case-2:

($\Theta_1 = \Theta_2$)



When the incidence angle (Θ_1) is equal to the critical angle, then the light ray co-incidences with the material boundary represents some loss of signal at the reception.

According to Snell's law

$$\boxed{n_1 \sin \Theta_1 = n_2 \sin \Theta_2}$$

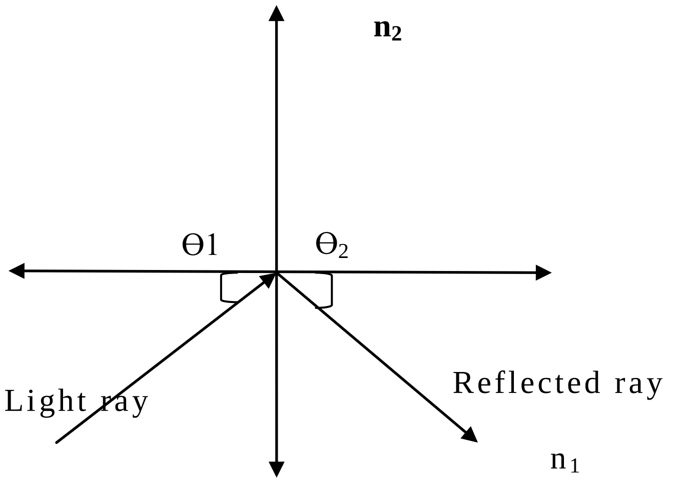
$\Theta_2 = 90^0$, so $\quad n_1 \sin \Theta_1 = n_2 \sin 90^0 \quad$ ($\sin 90^0 = 1$)

$\Theta_1 = \Theta_c$, $\quad n_1 \sin \Theta_c = n_2$

$$\boxed{\Theta_c = \sin^{-1}(n_2/n_1)}$$ Critical angle

Case-iii

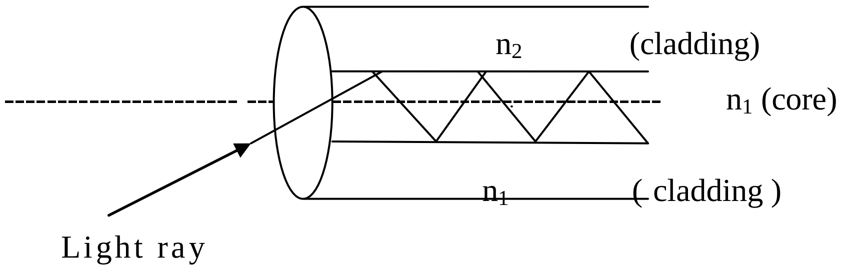
($\Theta_1 > \Theta_2$)



When the incident angle (Θ_1) is greater than the critical angle ($\Theta_1 > \Theta_c$) or the incident angle is equal to acceptance angle ($\Theta_1 = \Theta_A$) then the light ray is reflected back to the same medium. In this case, at the receiver side, we will get lossless signal. This is called total internal reflection.

Critical angle (or) Acceptance angle

The angle made by the light ray with the axis of fiber such that, the light ray will be propagated in the form of reflections throughout the core of the fiber.



Numerical Aperture

The ability of a fiber to accept the light signal is called as Numerical Aperture. The value of this is <1 and it ranges from 0.14 to 0.15. It is a dimension less parameter.

$$N.A.=\sqrt{n_1-n_2}$$

Where n_1=R.I. of core
n_2=R.I. of cladding

(or) N.A. $=n1\sqrt{2\,\Delta}$ $\Delta = n_1^2 - n_2^2 / 2n_1^2$

$$\boxed{\text{Acceptance angle } \Theta_A = \sin^{-1}(N.A)}$$

Advantages of Optical Fiber

1. Small in size, light in weight
2. Flexibility
3. Low noise, low losses
4. Transparency
5. Low cost
6. More secure in nature
7. More data rate
8. Large bandwidth (THz)
9. Reliability
10. Easy to maintain it.

Applications

1. In long distance telecommunications
2. Used in medicine (endoscope, burning of tumors, reducing eye sight)
3. Aircrafts
4. Under seas
5. Photographic
6. Radars

Modes and configurations of optical fiber

Mode:-
The propagation of light signal throughout the waveguide will be described in the form of EM waves is called as mode.

Based on the variations of the mode, fibers are two types

1. Step index fiber
2. Graded index fiber

There are available in single mode and multimode

Single mode fibers are used in long distance applications. Multimode fibers are used in short distance applications.

Multimode fibers have several advantages over single mode fibers.

1. Due to larger core diameter, it is very much easier to launch the optical power into the fiber.

2. It is very easy to join several numbers of fibers using splices and connectors.

3. LED's are sufficient to launch the optical power into the multimode fiber.

LED's have several advantages over LASER diodes

LED	LASER
1.They require less number of components to make	1.More complex circuit
2. Longer life time compared to LASER	2.Short life time
3. Used for short distance applications	3.Used for long distance applications